



BAHIRDAR INSTITUTION OF TECHNOLOGY

DEPARTMENT OF IS

OS INDIVIDUAL Project

Project title:

Arch Linux Installation ON VirtualBox

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1. Introduction

Background

An Operating System (OS) is the core software that manages computer hardware and provides services to users and applications. Modern computing environments increasingly rely on **virtualization technology**, which allows multiple operating systems to run on a single physical machine safely and efficiently.

Motivation

Installing operating systems directly on physical hardware may risk data loss and hardware incompatibility. Virtualization provides a **safe, controlled, and flexible environment** for learning, testing, and experimentation.

This project focuses on installing **Arch Linux** in a **virtual machine** using **Oracle VM VirtualBox**, ensuring the host Windows operating system remains unchanged.

2. Objectives

2.1 Objectives of Knowing Virtual Installation of Operating Systems

- To demonstrate understanding of OS installation procedures in a virtual environment.
- To apply theoretical OS concepts such as boot process, filesystems, and partitioning.
- To gain hands-on experience with Linux OS without additional hardware.
- To develop troubleshooting skills related to OS installation and configuration.
- To explore OS resource management (CPU, memory, storage) using virtual machines.
- To understand virtualization architecture and its importance in modern computing.
- To prepare learners for real-world system administration tasks.

2.2 Objectives of Knowing Worldwide PC Brands

- To identify major PC brands capable of running multiple operating systems.
- To understand hardware diversity and OS compatibility.
- To relate OS performance with hardware specifications.
- To appreciate industry standards affecting OS design.
- To recognize CPU architectures (Intel, AMD, ARM) used globally.

2.3 Objectives of Knowing Worldwide Mobile Phone Brands

- To identify major smartphone brands and their operating systems.

- To understand the relationship between mobile hardware and OS design.
- To analyze mobile OS ecosystems (Android, iOS).
- To understand Android fragmentation and OS portability.
- To relate hardware innovation to OS evolution.

3. Requirements

3.1 Hardware Requirements

- PC/Laptop (HP ZBook)
- Processor: Intel with Virtualization support
- RAM: Minimum 4 GB (Recommended 8 GB)
- Storage: At least 40 GB free space

3.2 Software Requirements

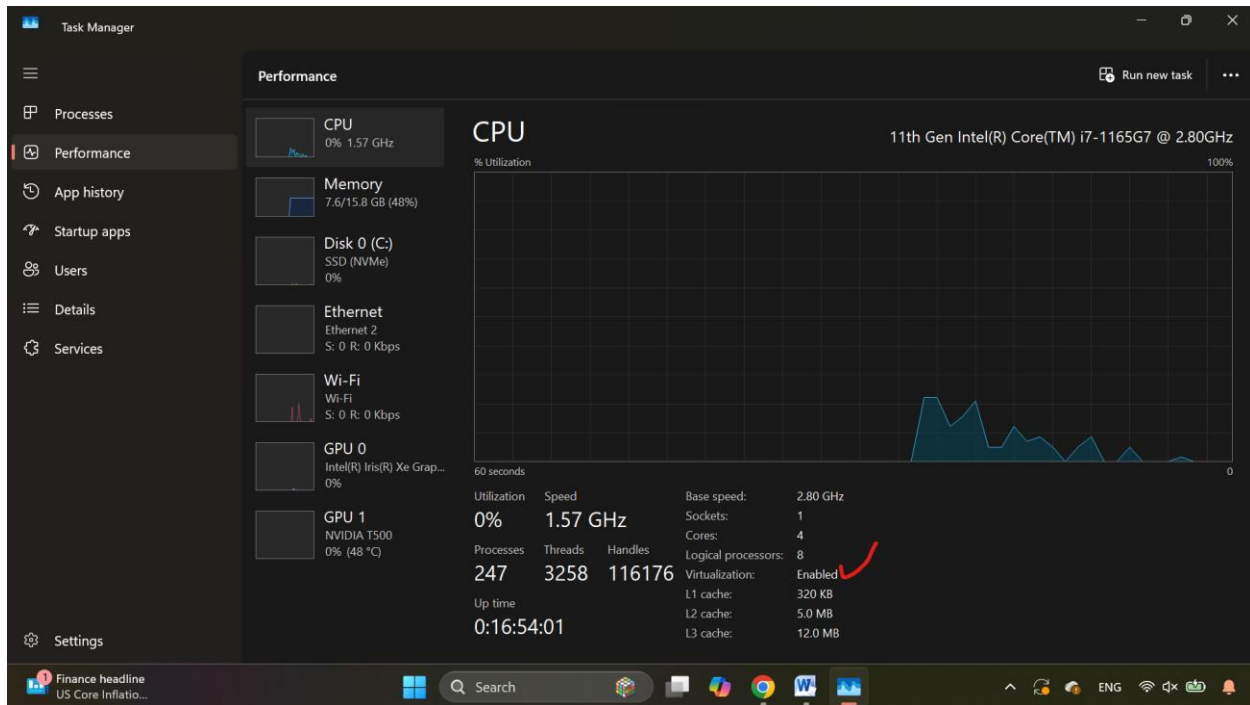
- Host OS: Windows 10/11
- Oracle VM VirtualBox
- Arch Linux ISO image
- Internet connection

4. Virtualization Tools Used

- **Oracle VM VirtualBox**
- (Other examples: VMware Workstation, Hyper-V)

5. Enable Virtualization (BIOS)

1. Restart the computer
2. Enter BIOS (F10 / F2 / Esc on HP)
3. Enable:
 - **Intel VT-x / AMD-V**
4. Save and exit

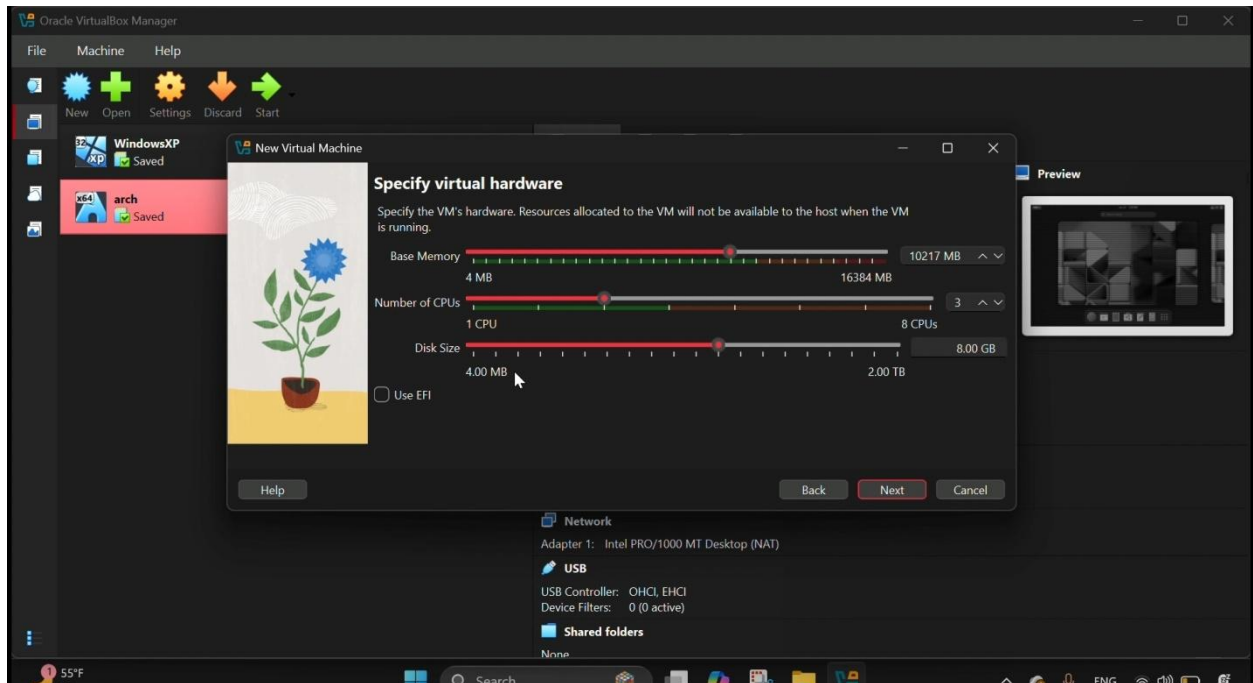


Without virtualization, VirtualBox will not run 64-bit systems.

6. Installation of Operating System in Virtual Environment

6.1 Virtual Machine Creation

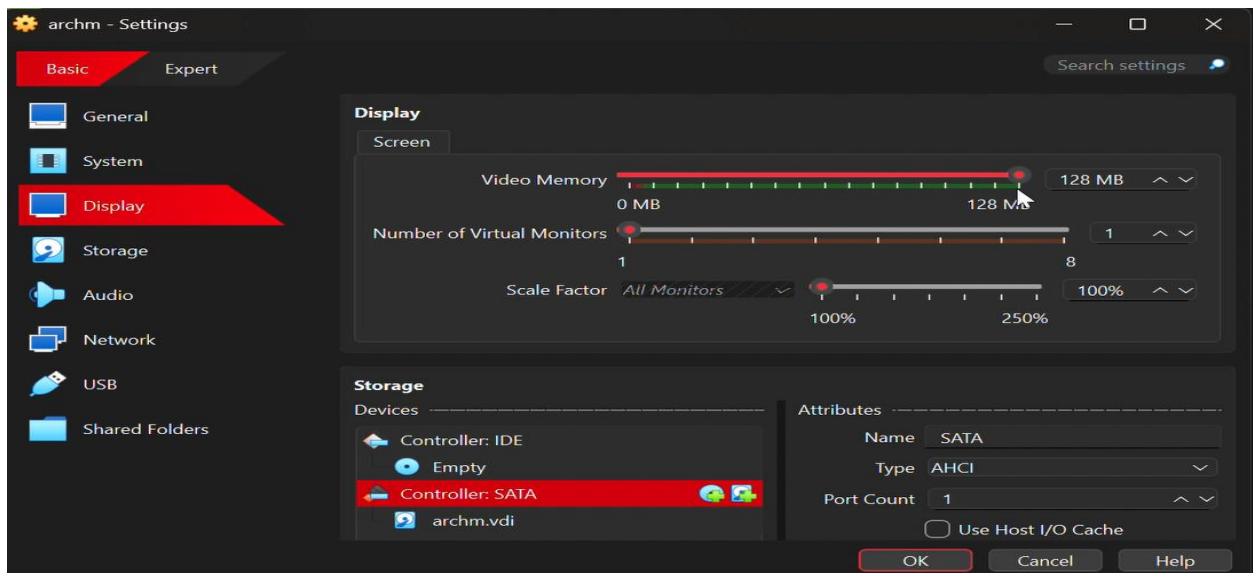
1. Open **Oracle VM VirtualBox**
2. Click **New**
3. Name: *Use your full name (e.g., Mogessie_Takele_ArchLinux)*
4. Type: Linux
5. Version: Arch Linux (64-bit)
6. RAM: 4096 MB
7. Disk: VDI → Dynamically allocated → 40 GB



1. Finish

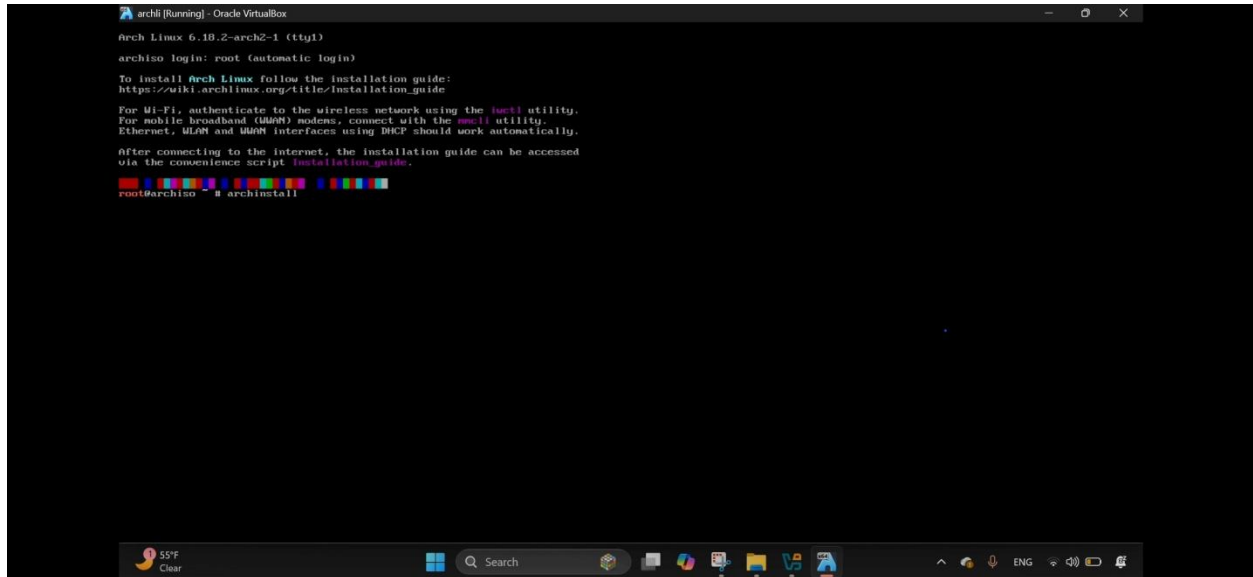
6.2. Attach Arch Linux ISO

1. VM → **Settings** → **Storage=120MB**
2. Controller IDE → Empty
3. Choose **Arch Linux ISO**
4. OK



7. Boot Arch Linux

1. Start VM
2. `archiso login: root`
3. Login automatically as root
4. Start installer:
5. `Archinstall`

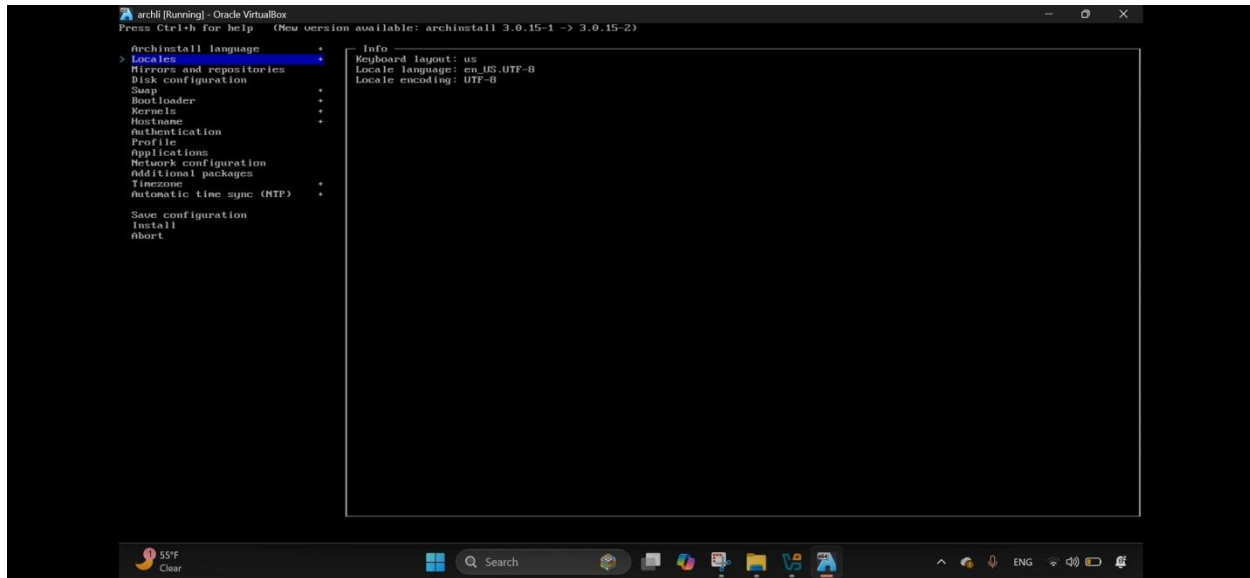


7. Archinstall Configuration

Archinstall Language

- Language: English
- Locale: en_US.UTF-8
- Keyboard: us

Locales



Mirrors and repositories

Select region=Germany

Disk Configuration

- Use best-effort default partition layout
- Disk: /dev/sda
- Wipe disk: **Yes**
- Partitions created automatically:

Mount	Size	FS
/boot	1 GiB	FAT32
/	Remaining	ext4

- Encryption: **Disabled**

Only the virtual disk is affected. Windows is safe.

Swap

- Swap on zram: **Disabled**

Bootloader

- **GRUB**

- Installed on `/dev/sda`

Kernel

- **linux (default)**

Hostname

- Example: `archlinux`

Authentication

- Root password: **Set**
- User created:
 - Username: `mogesie`

This step is mandatory. Installation will not proceed without it.

Profile

- Environment type: **desktop invironment**
- Selected: **GNOME**
- Packages installed include:
 - Gnome
 - Gnome-tweaks

Network Configuration

- **Use NetworkManager (iwid backend)**

Timezone & Time

- Region: **Africa**
- City: **Addis_Ababa**
- NTP: **Enabled**

8. Installation

1. Select **Install**
2. Confirm with **Yes**
3. System installs packages (10–20 minutes) based on internet connection


```

archli [Running] - Oracle VirtualBox
libarchive-3.8.5-1 libassuan-3.0.0-1 libbpf-1.6.2-1
libcap-2.77-1 libcap-ng-0.9-1 libelf-0.194-1
libevent-2.1.12-4 libffi-3.5.2-1 libgcrypt-1.11.2-1
libgpg-error-1.50-1 libidn2-2.3.0-1 libksba-1.6.7-2
libldap-2.6.10-2 libksmtp-1.0.1-1 libltdl-1.0.5-2
libnetfilter_conntrack-1.0.9-2 libnetfilter_conntrack-1.0.2-2
libnftnl-1.3.1-1 libnftnl-1.6.0-1 libnftnl-1.14.0-1
libnftnl-1.19.0-1 libnftnl-1.12.0-1 libnftnl-2.0.1-1
libnftnl-1.19.0-2 libnftnl-1.10.6-1 libnftnl-0.21.5-2
libnftnl-2.1.20-5 libnftnl-2.6.0-1 libnftnl-0.21.7-1
libnftnl-1.11.1-1 libnftnl-1.10.6-1 libnftnl-4.21.0-1
libnftnl-1.3.7-1 libnftnl-1.3-1 libnftnl-1.0.29-1
libnftnl-0.3.2-5 libnftnl-4.5.2-1 libnftnl-2.15.1-5
libnftnl-20240728-1 linux-api-headers-6.10-1
linux-firmware-andgpu-20260110-1
linux-firmware-atheros-20260110-1
linux-firmware-broadcom-20260110-1
linux-firmware-cirrus-20260110-1
linux-firmware-intel-20260110-1
linux-firmware-mediatek-20260110-1
linux-firmware-nvidia-20260110-1
linux-firmware-other-20260110-1
linux-firmware-radeon-20260110-1
linux-firmware-realtek-20260110-1
linux-firmware-silabs-20260110-1 lib-0.9.33-1 lz4-1.1.10.0-2
mkinitcpio-40-4 mkinitcpio-busybox-1.36.1-1 mpfr-4.2.2-1
ncurses-6.5-4 nettle-3.10.2-1 nft-1.0-1 openssl-3.6.0-1
pill-kit-0.25.10-2 pacman-7.1.0-77.0b174de-1
pacman-mirrorlist-20251021-1 pan-1.7.1-1 panbase-20250719-1
pciutils-3.14.0-1 pcre2-10.47-1 pinentry-1.3.2-2 post-1.19-2
procps-ng-4.0.5-3 psutils-3.7-1 readline-8.3.003-1 sed-4.9-3
shadow-4.10.0-1 sqlite-3.51.2-1 systemd-259-2
systemd-libs-259-2 systemd-sysucompat-259-2 tar-1.35-2
tpm2-tss-4.1.3-1 tzdata-2025-1 util-linux-2.41.3-2
util-linux-libs-2.41.3-2 xz-5.6.2-1 zlib-1.1.3.1-2
zstd-1.5.7-2 base-3-2 linux-6.10.5.arch1-1
linux-firmware-20260110-1 sudo-1.9.17.p2-2

Total Download Size: 654.03 MiB
Total Installed Size: 1167.05 MiB

:: Proceed with installation? [Y/n]
:: Retrieving packages...
linux-6.10.5.arch... 20.0 MiB 2.43 MiB/s 00:50 [###] 14%
linux-firmware-1... 5.6 MiB 650 KiB/s 02:11 [ ] 4%
linux-firmware-a... 5.6 MiB 713 KiB/s 02:13 [ ] 5%
linux-firmware-a... 3.2 MiB 318 KiB/s 02:20 [ ] 6%
gcc-libs-15.2.1-... 2.2 MiB 249 KiB/s 01:41 [ ] 6%
Total (0/141) 37.7 MiB 3.52 MiB/s 02:55 [ ] 5%

```

9. Installation Completed

Message shown:

Installation completed

Options:

- Exit archinstall
- Reboot system
- Chroot into installation

```

archli [Running] - Oracle VirtualBox
Press Ctrl+h for help

Installation completed in 13m33s
What would you like to do next?

> Exit archinstall
Reboot system
chroot into installation for post-installation configurations

```

✓ **Correct choice:**

Reboot system

10. Important Reboot Step

Before reboot:

- VirtualBox → Devices → Optical Drives → **Remove ISO**

Then reboot.

11. First Boot (Installed System)

```
archlinux login:
```

Login:

```
username: mogesie
password: ****
```

12. Issues (Problems Faced)

1. **Network unreachable**
2. **startx command not found**
3. **Pacman mirror errors**
4. **GUI not starting**
5. **Incorrect profile selection**

Problem	Solution
Network unreachable	Enable NetworkManager
startx not found	Install Xorg and Desktop Environment
Pacman errors	Update mirror list
GUI not loading	Install correct desktop profile
Login errors	Verify username/password

❑ `archinstall: ModuleNotFoundError`

Reason:

`archinstall` only exists in the live ISO.

☐ **Fix:**

Remove ISO before reboot.

I try to enter ☐ `startx: command not found`

Reason:

`xorg-xinit` missing or `.xinitrc` not configured.

☐ **Fix:**

`archinstall`

☐ **Network unreachable**

Reason:

`NetworkManager` not running.

☐ **Fix:**

- **Use NetworkManager (iwd backend)**

Check:

`ping archlinux.org`

13. Filesystem Support

Supported Filesystems

Filesystem	OS Support	Reason
NTFS	Windows	Large files, journaling
FAT32	Universal	Boot partitions
exFAT	Windows/Linux	Flash storage
ext4	Linux	Stability, performance
Btrfs	Linux	Snapshots
ZFS	Servers	Data integrity
APFS	macOS	Apple ecosystem

Chosen Filesystem: ext4

Why?

- Stable and reliable
- Best supported in Linux
- Low overhead
- Ideal for beginners

14. Advantages and Disadvantages

Advantages

- Safe installation without data loss
- Easy rollback and testing
- Supports learning multiple OSs
- Cost-effective

Disadvantages

- Lower performance than physical installation
 - Requires sufficient RAM/CPU
 - Limited hardware access
-

15. Virtualization in Modern Operating Systems

What is Virtualization?

Virtualization is a technology that allows multiple operating systems to run simultaneously on one physical machine.

Why Virtualization?

- Resource optimization
- Safe testing environment
- Cost efficiency
- Cloud computing foundation

How Virtualization Works

- Hypervisor manages hardware
- Virtual machines share resources

- Each VM runs isolated OS
-

16. Conclusion

This project successfully demonstrated the installation of **Arch Linux in a virtual environment** using **Oracle VM VirtualBox**. It enhanced understanding of operating systems, filesystems, virtualization, and troubleshooting. Virtualization proves to be a powerful and essential technology in modern computing.

17. Future Outlook / Recommendations

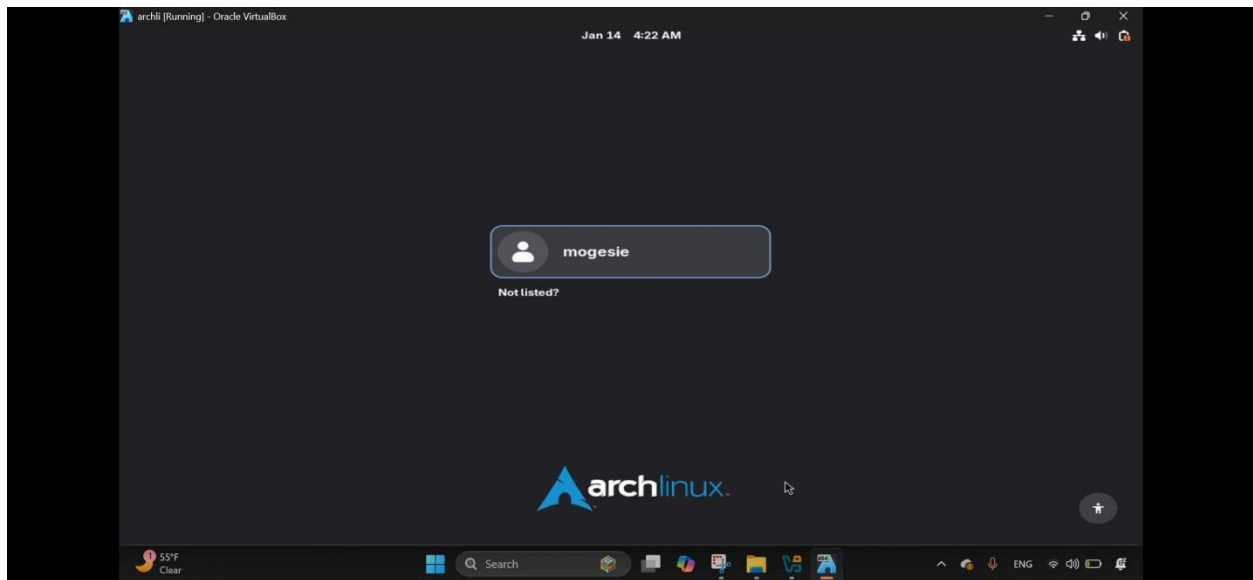
- Learn advanced Arch Linux system administration
 - Explore server deployment
 - Practice shell scripting
 - Study cloud virtualization platforms (AWS, Azure)
 - Experiment with different filesystems (Btrfs, ZFS)
-

18. Evaluation Reflection

- ✓ Demonstrated OS installation expertise
- ✓ Clear understanding of virtualization
- ✓ Proper filesystem selection
- ✓ Practical troubleshooting skills
- ✓ Confidence in technical execution

19. Final Result

- ✓ Arch Linux installed inside VirtualBox
- ✓ Windows system untouched
- ✓ Network working
- ✓ GUI (GNOME WM) working
- ✓ Stable Arch system



Installing Arch Linux proves strong understanding of **Linux system administration**.

Safe Installation – Windows Not Affected